

B_Sentiment_analyse

March 12, 2025

```
[3]: import pandas as pd
from easynmt import EasyNMT
from tqdm import tqdm; tqdm.pandas()
from ast import literal_eval
import numpy as np
from transformers import AutoTokenizer, AutoModelForSequenceClassification
import torch.nn.functional as F
from transformers import pipeline
import torch
```

```
[4]: df = pd.read_csv("df2.csv")
```

0.1 Vertaling

```
[7]: # Deze code is bedoeld om zinnen van de ene taal naar de andere te vertalen.
# Het vertaalmodel vertaalt zinnen van het Nederlands naar het Engels.
# De zinnen worden eerst in een lijst gezet zodat ze makkelijk verwerkt kunnen
    worden.
# Elke zin in de lijst wordt vertaald.
# De vertalingen van alle zinnen worden verzameld en teruggegeven.
```

```
model = EasyNMT("opus-mt", use_cuda=True)
model.device = 'cuda'

def get_translation(sentences, source_lang="nl", target_lang="en"):
    sentences = literal_eval(sentences)
    translations = []
    for s in sentences:
        translation = model.translate(s, source_lang=source_lang,
        target_lang=target_lang)
        translations.append(translation)
    return translations
```

```
[8]: # Deze code vertaalt de zinnen die het woord "POLICE" bevatten naar het Engels.
    ↪
```

```
df["en_police_sentences"] = df.police_sentences.progress_apply(lambda x:
↳get_translation(x))
```

```
0%|          | 0/5412 [00:00<?, ?it/s]/home/rdubel/.local/lib/python3.8/site-
packages/transformers/tokenization_utils_base.py:1601: FutureWarning:
`clean_up_tokenization_spaces` was not set. It will be set to `True` by default.
This behavior will be deprecated in transformers v4.45, and will be then set to
`False` by default. For more details check this issue:
https://github.com/huggingface/transformers/issues/31884
warnings.warn(
100%|         | 5412/5412 [58:17<00:00, 1.55it/s]
```

0.2 Sentiment berekenen

```
[15]: # Deze code haalt een model op dat sentimentanalyse kan uitvoeren, specifiek
↳gericht op aspecten in teksten.
```

```
absa_tokenizer = AutoTokenizer.from_pretrained("yangheng/
↳deberta-v3-base-absa-v1.1")
absa_model = AutoModelForSequenceClassification.from_pretrained("yangheng/
↳deberta-v3-base-absa-v1.1").to("cuda")
```

```
/home/rdubel/.local/lib/python3.8/site-
packages/transformers/tokenization_utils_base.py:1601: FutureWarning:
`clean_up_tokenization_spaces` was not set. It will be set to `True` by default.
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https://github.com/huggingface/transformers/issues/31884
warnings.warn(
/home/rdubel/.local/lib/python3.8/site-
packages/transformers/convert_slow_tokenizer.py:551: UserWarning: The
sentencepiece tokenizer that you are converting to a fast tokenizer uses the
byte fallback option which is not implemented in the fast tokenizers. In
practice this means that the fast version of the tokenizer can produce unknown
tokens whereas the sentencepiece version would have converted these unknown
tokens into a sequence of byte tokens matching the original piece of text.
warnings.warn(
```

```
[16]: # Het model bepaalt de waarschijnlijkheden voor drie mogelijke gevoelens:
↳negatief, neutraal en positief.
```

```
def get_aspect_based_sentiment(sentence, aspect="POLICE"):
    inputs = absa_tokenizer(f"[CLS] {sentence} [SEP] {aspect} [SEP]",
↳return_tensors="pt").to("cuda")
    outputs = absa_model(**inputs)
    probs = F.softmax(outputs.logits, dim=1)
    probs = probs.detach().cpu().numpy()[0]
    result = {"negative": probs[0], "neutral": probs[1], "positive": probs[2]}
```

```
return result
```

```
[17]: # Deze code analyseert het sentiment voor een lijst van zinnen, specifiek  
      ↳ gericht op het aspect "POLICE".
```

```
def get_sentence_sentiment(sentences):  
    results = []  
    sentences = literal_eval(sentences)  
    for sentence in sentences:  
        result = get_aspect_based_sentiment(sentence)  
        results.append(result)  
    return results
```

```
[18]: # Deze code berekent het sentiment voor de vertaalde zinnen die het woord  
      ↳ "POLICE" bevatten.
```

```
df_en["police_sentences_sentiment"] = df_en.en_police_sentences.  
    ↳progress_apply(lambda x: get_sentence_sentiment(x))
```

```
100%|          | 5412/5412 [05:21<00:00, 16.84it/s]
```

```
[20]: df_en.to_csv("df3.csv")
```